MicroPulse

HSR Intelligence — Coach Guide

Hamstring and soft-tissue injury monitoring for Lite Catapult plans

1. Overview — what is HSR Intelligence?

HSR Intelligence is a sport-science page in MicroPulse that monitors per-player hamstring and soft-tissue injury risk. It is built specifically for clubs on a Lite Catapult subscription (no B2-3 efforts exposed) but uses metrics every Catapult plan exposes: high-speed running, sprint distance, sprint count, and max velocity.

Think of it as the **Lite-tier equivalent of Decel Intelligence** — same evidence quality (BJSM-published research), different inputs. You get per-player red / yellow / green flags telling you who to hold back, who to push more, and who you can leave alone.

Key distinction

Decel Intelligence (McBurnie 2022) covers ACL, quadriceps and patellar tendon injuries — the ones caused by braking actions. HSR Intelligence covers hamstring and other soft-tissue strains — injuries that happen during the sprint itself, in the late-swing phase. The two pages tell you different things.

When do you see HSR Intelligence?

- Teams on a Lite Catapult plan see HSR Intelligence in the top-tab bar.
- Teams on a Pro / Premium plan see Decel Intelligence instead — higher fidelity because B2-3 efforts are available.
- If the Catapult tier is set to "unknown", MicroPulse auto-detects whether B2-3 data is present and selects the right page.

2. The science — three independent studies

HSR Intelligence combines three independent BJSM-published studies into one combined verdict. Each study contributes one sub-flag (red / yellow / green), and the system takes the worst of the three as the overall verdict.

2.1 — Malone et al. 2017 (HSR & Sprint ACWR)

Finding: Acute:Chronic Workload Ratio (ACWR) > 1.5 for high-speed running raises soft-tissue injury risk by **+200%** in elite soccer players. ACWR compares the 7-day acute load against the 28-day chronic load (per-day average).

De-training paradox: ACWR < 0.8 with a high chronic load (>200 m/day average) raises risk by +130%. This is the classic "underloading" trap — a player who missed weeks of training but comes back with low acute volume has lost the protection that chronic load provides.

2.2 — Buchheit & Mendez-Villanueva 2014 (Max-V Exposure)

Finding: Players who reach **≥95% of their personal max velocity regularly** suffer **FEWER** hamstring strains. Players who never approach their own max-V are at higher risk when they're forced to (e.g. in a competitive match) — the muscle isn't accustomed to the full-speed demand.

MicroPulse measures this with two numbers: %MaxV (best velocity reading in the last 28 days / personal all-time max × 100) and the count of sessions where the player reached ≥95% of their own max-V in the last 28 days.

2.3 — Bowen et al. 2017 (chronic-load buffer)

Finding: High chronic load + low acute = protected. High chronic + high acute = highest risk. Low chronic + high acute = also high risk. HSR Intelligence applies this by *not* flagging low-acute in a low-chronic window (e.g. early pre-season when the team is just starting up) as risk — that would be a false positive.

3. Column-by-column glossary

When you open HSR Intelligence you see a table with one player per row, sorted by risk severity (red → green). Here is exactly what each column shows:

Column	What it measures	How to read it
Player	Player name and position (defender / midfielder / forward).	Self-explanatory — no calculation.
Days 7/28	Number of days with GPS data in the 7-day and 28-day windows.	If Days 28 < 5, the player gets 'Need more data'. You need at least 1 week of data.
HSR 7d (m)	Total high-speed running over the past 7 days (metres). HSR = distance covered above ~19.8 km/h.	Compare across players: who has accumulated the most load this week.
HSR ACWR	Acute:Chronic ratio for HSR. (HSR 7d / 7) ÷ (HSR 28d / 28).	This is the headline number on the page. >1.5 = red, 1.3-1.5 = yellow, <0.8 with high chronic = de-training yellow.
Sprint 7d (m)	Total sprint distance over the past 7 days (metres). Sprint = distance covered above ~25.2 km/h.	Secondary signal — sprint distance and HSR usually move together but not always.
Sprint efforts 7d (#)	Number of discrete sprint efforts in the last 7 days (Velocity Band 6 efforts).	Each sprint is a discrete event. 5-15 per week is typical for working players; 20+ is high.
Sprint ACWR	Same formula as HSR ACWR but built on sprint effort count. >1.5 = red, 1.3-1.5 = yellow.	A sprint-effort spike most often happens after a player plays an intense match following a week off.
%MaxV (28d)	Best velocity in the last 28 days as a percentage of personal all-time max (last 365 days).	<90% = red if 0 sessions over 95%. 90-92% with <2 sessions = yellow. >92% or 2+ sessions = green.
≥95% MaxV sessions	Number of sessions in the last 28 days where the player reached ≥95% of their own max-V.	Target: 4+ in 28 days (≈1 per week) — Buchheit's protective dose.
Status	Combined verdict — worst of the three flags (HSR ACWR, Sprint ACWR, MaxV).	This is what you read first. The pill is colour-coded.

4. Colour codes — what the flags mean

Every number in the table is self-colour-coded by its own sub-flag. The rightmost column (Status) is worst-of-three — that is the flag you act on.

Status	What it means	What you should do
High risk	At least one of three signals is red. Risk is measurable and significant.	Look at which column is red (HSR ACWR, Sprint ACWR, or %MaxV). Adjust load accordingly.
Watch	No red flags but at least one signal is in the yellow band. Not acute risk, but trending that way.	Priority-monitor the next 2-3 days. You can continue training but with closer attention to RPE and soreness.
Healthy	All three signals are green. Load is within normal range and max-V exposure is sufficient.	No specific action — train normally.
Need more data	Fewer than 5 days of data in the 28-day window, or personal max-V is missing.	Sync Catapult / upload CSV for two weeks before trusting the output.

5. Which sub-flag is red — and what does it mean?

The overall status pill tells you the risk level, but you need to read the sub-flags to know what to do. Here is what each combination means:

Scenario A: HSR ACWR > 1.5 (red on HSR numbers)

What's happening: The last 7 days had 50%+ more HSR than the 28-day average. Classic spike pattern — player did two hard sessions in a row, or played two games in 4 days, or returned from rest straight into an HIT block.

Action: Put the next session on low-HSR (recovery, technical work, scheme running). Avoid HIT blocks, Yo-Yo, or Wattbike running drills for 2-3 days. Let chronic load catch up before the next spike.

Scenario B: HSR ACWR < 0.8 with high chronic load (yellow — de-training paradox)

What's happening: Player has missed sessions (injury, illness, a deload week that dragged on), but chronic load is still high (>200 m/day average). They are losing the protection chronic load provided and are vulnerable when they return.

Action: Re-ramp slowly. Their first session back should be 50-60% of typical HSR — not straight into a full session. Use the MicroPulse Quadrant page to see total load (not just HSR).

Scenario C: Sprint ACWR > 1.5 (red on Sprint numbers)

What's happening: Spike in sprint effort count. Most often after a match where the player was forced into many defensive sprints, or after an HIT session that ran long. Each sprint is a discrete event that micro-damages muscle.

Action: Plan the next 48 hours as a sprint-arm (no SSGs with goal-scoring opportunities that pull players into real sprints). Focus on possession and technical work in this window.

Scenario D: %MaxV < 90% with 0 sessions ≥95% (red on MaxV numbers)

What's happening: Player has not approached their own max velocity in 28 days. This is a *different kind of risk* — not spike injury, but acute-demand injury. The muscle isn't used to full-speed effort and tears when forced to (e.g. in a match where the player needs a 30+ metre sprint).

Action: This is the **opposite of the other scenarios** — you need to give the player MORE sprinting, not less. Add 1-2 acceleration drills per week (10-30 m sprints with full recovery). Target: ≥1 session per week where they reach ≥95% of their own max-V (Buchheit's protective dose).

> **Important distinction**
> Many coaches assume that lower max-V is always safer. It is not. Buchheit 2014 shows that players who never approach their own max-V are at higher risk when forced to. Protection is a function of regular high-speed exposure, not avoidance of it.

6. Workflow — when to use the page

Daily — before training (5 min)

- Open HSR Intelligence after Catapult data syncs (typically the morning after the previous evening's session / match).
- Read the red rows first — which players need an adjustment today?
- Read which sub-flag is red — an HSR spike calls for a different action than MaxV under-exposure.
- Adjust the plan in Week Setup or tell individual players to modify intensity.

Weekly — match-week setup (10 min)

- After the match: review HSR ACWR for every player — who is now in the red zone?
- Before planning next week: check who is in yellow (de-training paradox) — they need ramp-up, not recovery.
- Cross-check %MaxV — are any players losing max-V exposure? Plan an acceleration block for them.

Monthly — injury review (15 min)

- Compare who was red the week before each recent hamstring injury — is there correlation? (Use Injury Log + correlation engine.)
- Look at who has been yellow for an extended period (4+ weeks) — chronic monitoring case.
- Discuss with S&C — is the programming hitting ≥1 max-V session per week per player?

7. Frequently asked questions

Why does one player show "Need more data" while others don't?

The page needs at least 5 days of GPS data in the last 28 days to compute a reliable ACWR. It also needs personal max-V (from the last 365 days) to compute %MaxV. New players, or players returning from injury, may need 1-2 weeks of data before the system trusts the output.

What if a player is red but says they feel fine?

That's the point — injuries arrive before symptoms. Malone 2017 showed that spike patterns are predictive 7-14 days before the injury manifests. You should act before the player reports feeling tired, not after.

Why is %MaxV measured over 28 days and not 7?

Max velocity is a rare event — a player reaches their own max-V perhaps 1-2 times per week, not every session. A 7-day window would miss many real max-V sessions. 28 days is long enough to capture the protective-dose measurement but not so long that data from inactive periods skews it.

What if I don't see HSR Intelligence in the top-tab bar?

Your team is probably on a Pro / Premium plan. In that case you'll see Decel Intelligence instead — a higher-fidelity sport-science page built on B2-3 efforts. If you want to see both you can change the plan tier in Settings → Catapult → Plan Type.

Can I tune these thresholds?

No — the thresholds are taken directly from the Malone 2017 and Buchheit 2014 papers. Changing them would lose the page's evidence base. If you want lower (sensitive) or higher (specific) warnings, that's what Quadrant View and Foster Strain are for — broader overviews.

8. Limitations — what HSR Intelligence CAN'T do

It is important to know the system's limitations so you use it correctly and don't trust it for things it doesn't measure.

- This is a Lite-tier page. It cannot identify the deceleration burden (B2-3 efforts are missing) — that requires a Pro plan + Decel Intelligence.
- It does not detect knee injuries (ACL / quad / patellar tendon) — that is a decel mechanism this dataset doesn't capture.
- It does not account for subjective wellness (sleep, soreness, mood). Use the Decision Summary alongside (Saw 2016: subjective wellness > objective markers).
- It does not detect single high-load efforts in critical match minutes (e.g. final 15 min when injuries most often occur) — only daily aggregation.
- It does not see non-Catapult activity (gym, personal training, other-club sessions). Trust the player's RPE entries too.

9. References

Malone S, Owen A, Mendes B, Hughes B, Collins K, Gabbett TJ. (2017). The acute:chronic workload ratio in relation to injury risk in professional soccer. *Journal of Science and Medicine in Sport*, 20(6), 561-565.

Buchheit M, Mendez-Villanueva A. (2014). Effects of age, maturity and body dimensions on match running performance in highly trained under-15 soccer players. *Journal of Sports Sciences*, 32(13), 1271-1278.

Bowen L, Gross AS, Gimpel M, Li FX. (2017). Accumulated workloads and the acute:chronic workload ratio relate to injury risk in elite youth football players. *British Journal of Sports Medicine*, 51(5), 452-459.

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